# Lecture 10

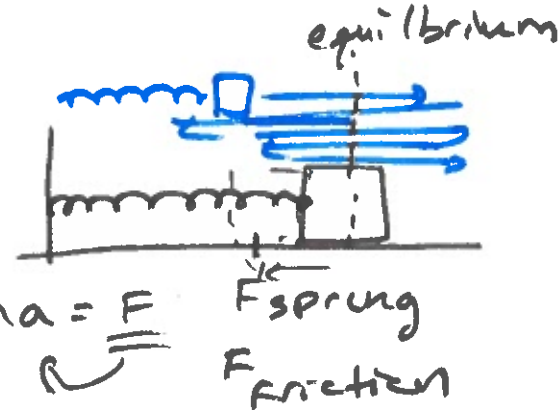
## The Mass-Spring Oscillator

position $y$
velocity $v = y'$
acceleration $a = v' = y''$

As we have seen previously, Newton's Second Law ($F = ma$) can be used to construct differential equations. A first order equation can be created in this way by using the fact that acceleration is the first derivative of velocity.

$m \frac{dv}{dt} = F$ 1st order in velocity

If instead, we consider the scenario where acceleration is the second derivative of position, then we can construct second order equations using Newton's Second Law.




One of the most straightforward physical scenarios to model this way is the mass-spring oscillator. If we supposed that a mass $m$ is attached to a spring fixed at one end, then we get the following equation

$$my'' + by' + ky = F_{\text{ext}} \tag{10.1}$$

← external forcing function.

ma  friction  spring

where

- $y$ is displacement from equilibrium
- $m$ is the mass of the object attached to the spring
- $k$ is the spring constant (stiffness), from Hooke's Law  $F_{spring} = -ky$
- $b$ is the damping coefficient (usually friction, but some force opposing the motion of the spring)  $F_{friction} = -by'$
- $F_{\text{ext}}$ is the sum of all external forces.

The motion of an object under these conditions may oscillate around the equilibrium (rest) and can be modeled with a trigonometric function. This happens if we have no damping coefficient and no external forces, i.e. $b = 0$ and $F_{\text{ext}} = 0$.

cos or sin

In this case (10.1) resolves to

$$my'' + \overset{0}{b}y' + ky = \overset{0}{F_{ext}}$$

$$my'' + ky = 0$$

For such an equation we expect the solution to be a cosine or sine function.

**Example:** Verify that $y(t) = \cos(\omega t)$ where $\omega$ is the angular frequency, depending upon $m$ and $k$, is a solution to

$$my'' + ky = 0$$

$$\left.\begin{aligned} y &= \cos(\omega t) \\ y' &= -\sin(\omega t)\cdot\omega \\ y'' &= -\cos(\omega t)\,\omega^2 \end{aligned}\right\} \text{substitute}$$

$$m(-\cos(\omega t)\omega^2) + k\cos(\omega t) \overset{?}{=} 0$$

$$\Rightarrow \cos(\omega t)[k - m\omega^2] \overset{?}{=} 0$$

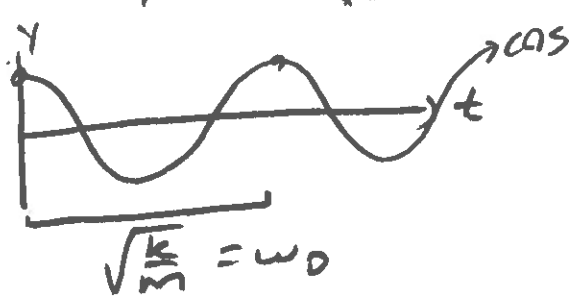
(true for all values of $t$).

$$\Rightarrow k - m\omega^2 = 0$$

$$\Rightarrow m\omega^2 = k \Rightarrow \omega^2 = \frac{k}{m}$$

$$\Rightarrow \omega = \sqrt{\frac{k}{m}} = \omega_0$$ frequency of the spring

So $y = \cos\left(\sqrt{\frac{k}{m}}\, t\right)$ is a solution to DE.




$m \uparrow \quad \omega_0 \downarrow$ slower

$k \uparrow \quad \omega_0 \uparrow$ faster

$my'' + by' + ky = 0$

If we introduce damping ($b \neq 0$) then the object will not oscillate perpetually and will eventually come to rest at equilibrium. Depending upon the value of $b$ relative to $m$ and $k$ we can have a faster or slower level of damping. In certain scenarios these can both be advantageous. For instance, a car suspension uses springs to dampen the oscillations caused by deviations in the road surface and we desire those oscillations to die off quickly. Whereas, a tuning fork vibrates for some time and its oscillations die off more slowly.

$y$

$ax^2 + bx + c = 0$

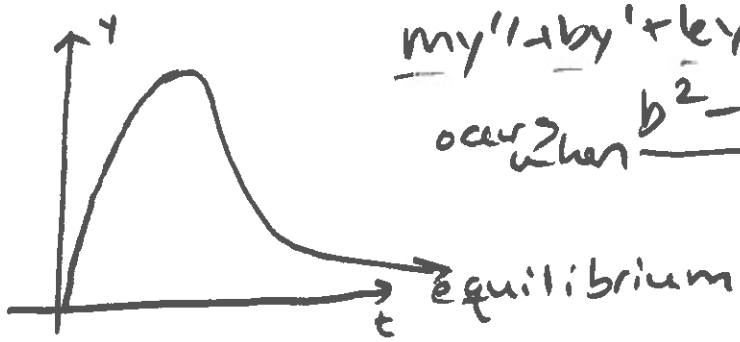
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$b^2 - 4ac$ discriminant

The car suspension is an example of an **overdamped** system

$my'' + by' + ky = 0$

occurs when $b^2 - 4mk > 0$



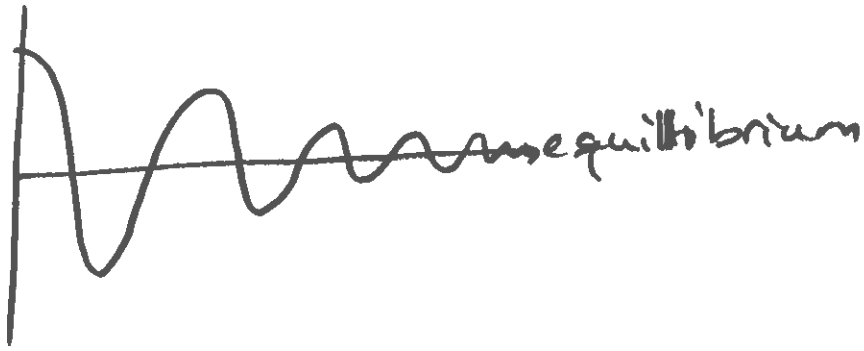

friction — $b^2 - 4mk > 0$ overdamped

$b^2 - 4mk = 0$

$b^2 - 4mk < 0$ underdamped

The tuning fork is an example of an **underdamped** system

$b^2 - 4mk < 0$



**Example:** Show that $y(t) = te^{-t}$ is a solution to the mass-spring equation (10.1) if $F_{\text{ext}} = 0$, $m = 1$, $b = 2$, and $k = 1$

$$my'' + by' + ky = F_{ext}$$

$$\Rightarrow \quad y'' + 2y' + y = 0$$

so $y = te^{-t}$ is a solution

$$y' = e^{-t} + t(-e^{-t}) = \underline{(1-t)e^{-t}}$$

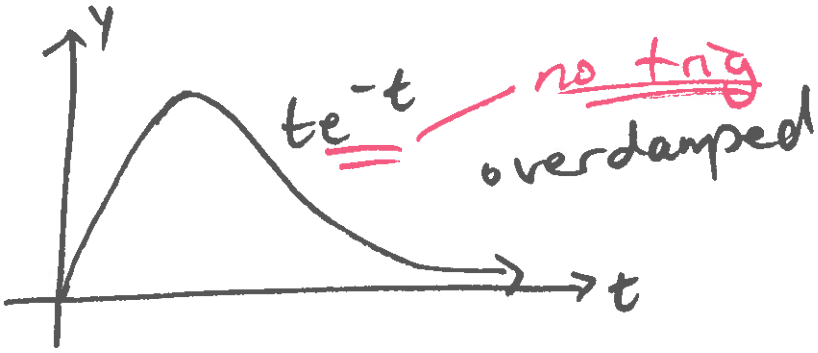
$$y'' = (-1)e^{-t} + (1-t)(-e^{-t}) = (t-2)e^{-t}$$

$$y'' + 2y' + y = \underbrace{(t-2)e^{-t}}_{y''} + 2\underbrace{((1-t)e^{-t})}_{y'} + \underbrace{te^{-t}}_{y}$$

$$= te^{-t} - \cancel{2e^{-t}} + \cancel{2e^{-t}} - 2te^{-t} + te^{-t}$$

$$= 0 \checkmark \quad \text{solves the equation.}$$

$$b^2 - 4mk = 2^2 - 4(1)(1) = 4 - 4 = 0 \; ?$$



**Example:** Show that $y(t) = e^{-t}\cos(t)$ is a solution to the mass-spring equation (10.1) if $F_{\text{ext}} = 0$, $m = 1$, $b = 2$, $k = 2$.

$$b^2 - 4mk = 2^2 - 4(1)(2)$$
$$= 4 - 8 = -4 < 0$$
underdamped

$$my'' + by' + ky = F_{ext}$$

$$\Rightarrow y'' + 2y' + 2y = 0$$

$$y = e^{-t}\cos(t), \quad y' = -e^{-t}\cos(t) + e^{-t}(-\sin(t))$$

$$y'' = [e^{-t}\cos(t) - e^{-t}(-\sin(t))] + [-e^{-t}(-\sin(t)) + e^{-t}(-\cos(t))]$$

$$= \cancel{e^{-t}\cos(t)} + e^{-t}\sin(t) + e^{-t}\sin(t) - \cancel{e^{-t}\cos(t)}$$

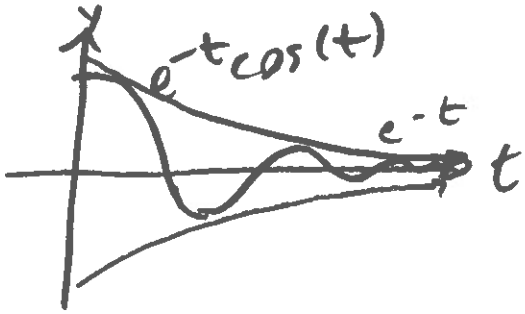
$$= 2e^{-t}\sin(t)$$

$$y'' + 2y' + 2y = [\underbrace{2e^{-t}\sin(t)}_{y''}] + 2[\underbrace{-e^{-t}\cos(t) - e^{-t}\sin(t)}_{y'}] + 2(\underbrace{e^{-t}\cos(t)}_{y})$$

$$= \cancel{2e^{-t}\sin(t)} - \cancel{2e^{-t}\cos(t)} - \cancel{2e^{-t}\sin(t)} + \cancel{2e^{-t}\cos(t)}$$

$= 0$ ✓ satisfies the DE.

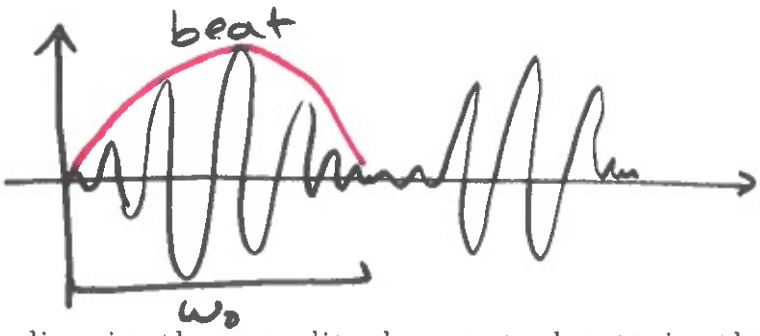
underdamped



When we begin to introduce external forces ($F_{\text{ext}} \neq 0$) that are sinusoidal in nature, then the behavior of the object oscillating can start off erratic but relax into another oscillation, perhaps at a different frequency. Some examples of this behavior are speakers, ocean tides, and circuit amplifiers. Some systems can become destructive when external forces are applied through a type of response called resonance.

If we consider the undamped case ($b = 0$) and impose an external force at a frequency, $\omega$ close to that of the system, $\omega_0$ then we obtain oscillations such as

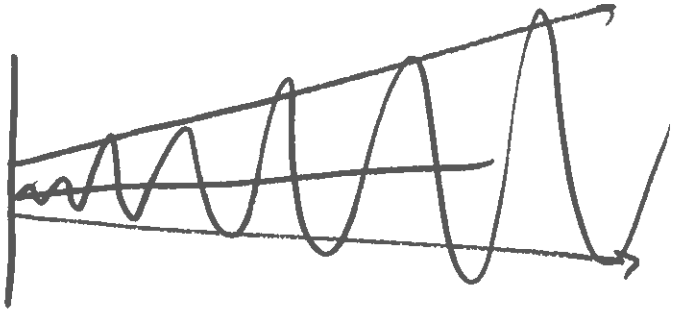
$$\underbrace{my'' + ky}_{\omega_0 = \sqrt{\frac{k}{m}}} = F_{ext} \sin(\omega t) \leftarrow \text{frequency } \omega \approx \omega_0$$




These dips in the amplitude create beats in the oscillations. For example, when a stringed instrument is tuned using a reference sound these beats occur. Adjusting the frequency (tension) in the string can quicken or slow down the beat frequency.

However, if the external force has a frequency **equal** to the spring's frequency $\omega_0$ then **resonance** occurs.

$$\omega = \omega_0 = \sqrt{\frac{k}{m}}$$



destructive resonance

no friction

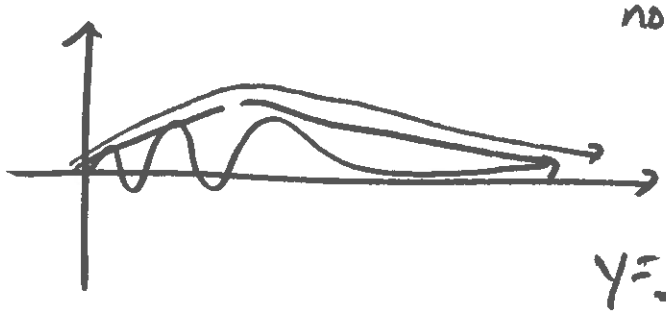
$y'' + y = \sin(t)$

$y \approx t\sin(t)$

friction

nondestructive resonance

resonance



$y'' + 2y' + 2y = e^{-t}\cos(t)$

$y = te^{-t}\cos(t)$

friction

Resonance can be harmful to physical structures such as bridges or buildings which have oscillating movement due to wind or other natural phenomenon. It can be useful in other applications such as

- Seismographs use resonance to amplify tremors in the earth
- Fluorescent lighting
- Radio tuning; aligning the frequency amplifies the signal
- Microwave ovens

In addition to the mass-spring oscillator, other oscillating systems occur in a number of physical scenarios. Some examples include:

**Rotational Mechanics**:

$$I\frac{d^2\theta}{dt^2} + \Gamma\frac{d\theta}{dt} + \mu\theta = \tau$$

where $I$ is the moment of inertia, $\mu$ is the torsion constant, $\Gamma$ is the rotational friction, $\tau$ is the drive torque, and $\theta$ is the rotational angle.

**Series RLC Circuit**:

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = e$$

where $L$ is the inductance, $q$ is the charge, $C$ is the capacitance, $R$ is the resistance, and $e$ is the voltage.